

Complete on the Finite Horizon: Soundness, Completeness, and Counterstrategy Trees for the Belief Recursion

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Abstract

For sufficient certificates the obstruction calculus has been honest about an open middle ground: its certificates prove nonviability, and their absence proves nothing. This paper, the D5 item of the generalization catalogue, closes that gap on the finite horizon for finite systems: the backward belief-safety recursion $W_0 = \{B \subseteq \mathcal{V}_T\}$, $W_{k+1} = \text{Pre}(W_k)$ is proved sound *and complete* for finite-horizon observation-based viability, nonviability is certified by a finite counterstrategy tree extracted constructively from the failed recursion, and on the programme’s two-patch working system the stabilized recursion coincides exactly with the successor’s $H/H - 2$ stability convention — the convention becomes a theorem. The audited family is horizon-free (verdicts at $N = 2, 4, 8$ coincide for every belief and structure), and every belief outside the stable kernel admits a validated tree: on finite systems the certificate gap of the calculus is empty. All claims are machine-verified in exact arithmetic (6/6 check families).

Introduction and relation to the programme

The programme’s papers state sufficient certificates: adverse drift, common-action emptiness, timing deadlines, certification limits. The source catalogue’s proposed strengthening is adopted here: on finite systems, finite-horizon completeness — the backward recursion is not merely sound but *equal* to the truth, and failure carries its own finite certificate. The relation to the completed items is precise: the viable selector’s recursive correspondence provides the semantics; the successor’s harness convention ($H = 12$ with an $H - 2$ stability check) is justified here as the stabilization of the recursion rather than assumed.

The backward recursion

Work on the two-patch system (16 states, safe set $\mathcal{V} = \{z_1 \geq 1, z_2 \geq 1\}$, one instrument per step, per-cell read-then-act semantics). Beliefs are finite subsets of $\mathcal{V}$. Define

$$W_0 = \{B : B \subseteq \mathcal{V}\},$$

$W_{k+1} = \{B : \text{some per-cell action tuple keeps every branch in } B\}$

Proposition 1 (Monotone decrease). $W_{k+1} \subseteq W_k$ for every k .

Proof. W_0 contains every safe belief. If $B \in W_{k+1}$ then some action tuple sends every successor part into W_k ; since each such part is safe, B is safe, so $B \in W_0$; and by induction along the same action tuple the parts lie in each earlier family, so $B \in W_k$. $\square$

Theorem 1 (Soundness and completeness on the finite horizon). For every belief B , information structure, and horizon N : $B \in W_N$ if and only if the regulator wins the N -step observation-based viability game from B . Consequently, if $B \notin W_N$ the recursion yields a finite counterstrategy tree of depth at most N whose leaves are forced exits of $\mathcal{V}$.

Proof. Induction on N . For $N = 0$ both sides read “ $B \subseteq \mathcal{V}$.” For $N + 1$: if $B \in W_{N+1}$, the witnessing action tuple followed by the winning strategies of the induction hypothesis on each successor part wins the game. Conversely, if the regulator wins at $N + 1$, the first move is a per-cell action tuple all of whose adversary responses leave the corresponding parts winning at N , i.e. in W_N by induction; hence $B \in W_{N+1}$. For the tree: if $B \notin W_{N+1}$, then for *every* action tuple some response reaches a part outside W_N ; the adversary branches over all tuples, selecting such a part each time; the construction terminates at depth N with leaves outside W_0 , i.e. forced exits. Finiteness of the state, action, and observation spaces makes the tree finite. $\square$

Corollary 1 (The stable family is the kernel; the $H/H - 2$ convention). On a finite system the chain $W_0 \supseteq W_1 \supseteq \dots$ stabilizes. Its stable family equals the infinite-horizon observation-based kernel, and membership in it coincides with the successor harness’s convention (verdict at H agreeing with the verdict at $H - 2$).

Proof. Monotone decrease in a finite lattice forces stabilization. After stabilization the recursion is a fixed point of the belief predecessor operator, i.e. the greatest fixed point below W_0 — the kernel. The two-step coincidence of verdicts is exactly two stabilizer iterations. $\square$

Verified record (6/6 exact check families)

E1. The recursion is computed as explicit belief families over all 2^9 safe subsets, under three structures (blind, aggregate, full); $W_{k+1} \subseteq W_k$ holds setwise at every step.

E2. Soundness and completeness agree instance by instance: for all 37 audited beliefs (36 safe pairs plus the three-state audited belief), 3 structures, and $N = 1, \dots, 6$: membership in

W_N equals the forward game value — 666 identifications, no exceptions.

E3. The belief $\{(1, 2), (2, 1)\}$ under the aggregate reading lies outside W_2 ; the constructive proof extracts its counterstrategy tree and every leaf validates as a forced exit.

E4. Each of the three chains stabilizes, and the stable family agrees exactly with the $H = 12/H - 2 = 10$ verdicts of the successor harness on all audited beliefs — the convention is a theorem here.

E5 (horizon-freeness, an exact finding). The planned search for a horizon-necessity instance (viable at $N = 2$, nonviable at $N = 4$) found none: verdicts at $N = 2, 4, 8$ coincide for every audited belief and structure. On this system viability is decided by depth 2; the finding is recorded rather than forced.

E6 (no asymptotic gap). Every belief outside the stable kernel admits a validated finite counterstrategy tree at the stabilization depth: on finite systems, nonviability is never merely witnessed — it is certified.

What is not claimed

Completeness is a finite-horizon, finite-system statement: the infinite-horizon calculus for general systems keeps its certificate gap by design, and nothing here closes it. Continuous state spaces, stochastic transitions (the stochastic selector’s setting), and hybrid mode switching (the programme’s hybrid companion) are out of scope; the counterstrategy tree is a game object, not a probabilistic witness. No computational complexity claims are made: the audited systems are small, and the recursion’s practical cost is acknowledged.

Declarations

Funding. None. **Competing interests.** None. **Data availability.** No external data were used. **Code availability.** The verification script `finite_horizon_completeness_v1_verify.py` is publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); it reproduces all six check families verbatim. **AI declaration.** GLM (Z.ai), Qwen (Alibaba Cloud) and DeepSeek AI assisted with drafting and iterative review.